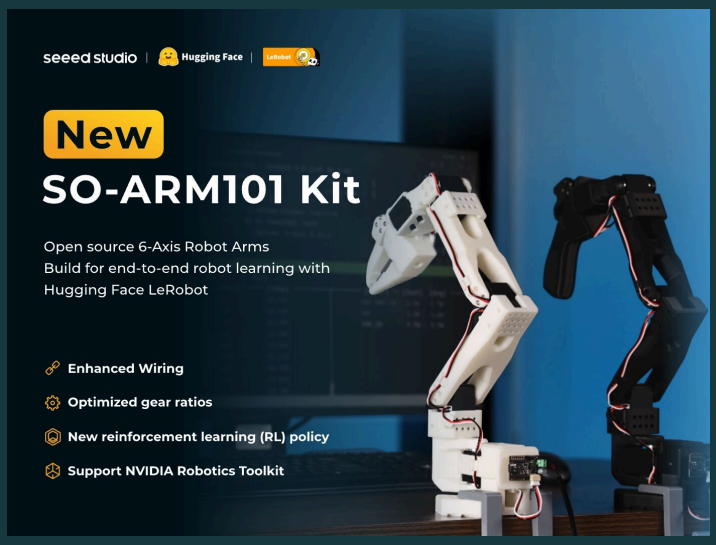
seeed studio

INDUSTRIAL PRODUCT DATASHEET

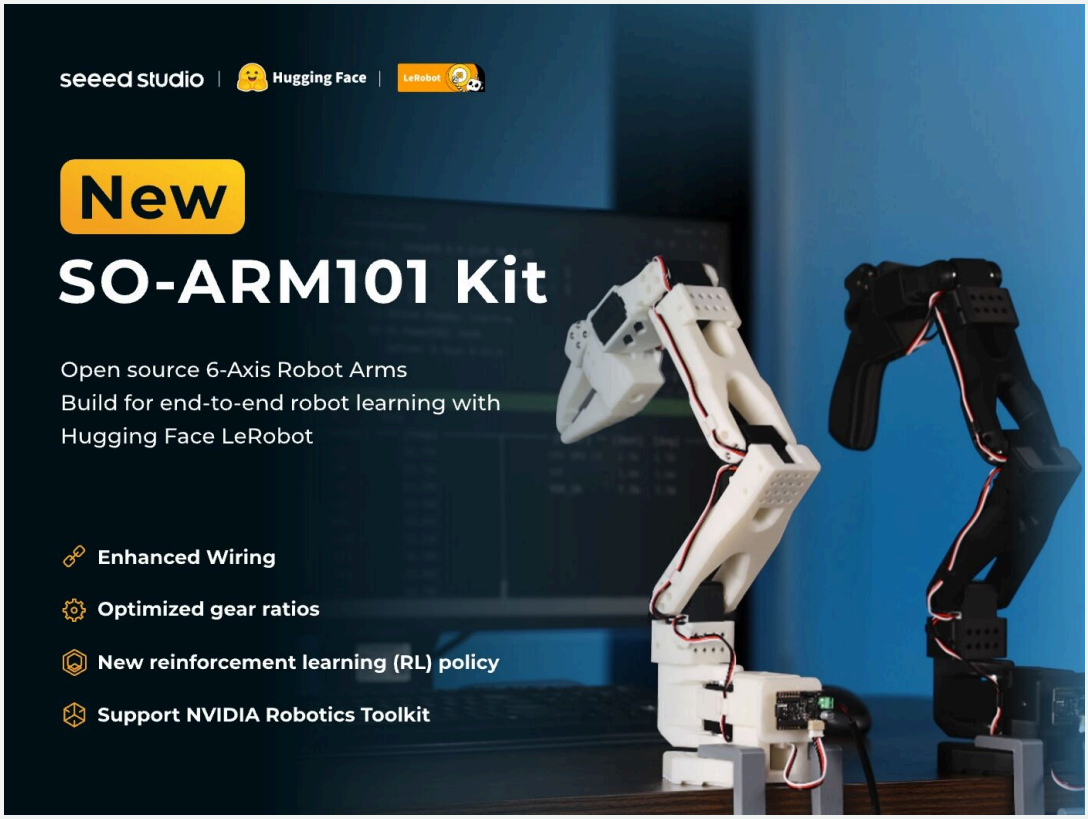
[New] SO-ARM101 Low-Cost AI Arm Servo Motor Kit for LeRobot (Without 3D Printed Parts)

[New] SO-ARM101 Low-Cost AI Arm Servo Motor Kit for LeRobot (Without 3D Printed Parts) is a 6-DOF open-source robotic arm with 30 kg·cm per joint servos, native Hugging Face LeRobot support, and SenseCraft Robotics guided...

SKU	UPDATED	PRICE
114993666	2026-09-14 02:49:20	$240.00(Excl. VAT)



Overview



The [New] SO-ARM101 Low-Cost AI Arm Servo Motor Kit for LeRobot (Without 3D Printed Parts) is a 6-DOF robotic arm with six STS3215 magnetic-encoder bus servos delivering 30 kg·cm torque per joint and 12-bit absolute position feedback. It provides native Hugging Face LeRobot support for imitation learning and reinforcement learning, along with fully open-source hardware/software and SenseCraft Robotics guided setup for cloud training and validation on NVIDIA Jetson devices.

Key Features

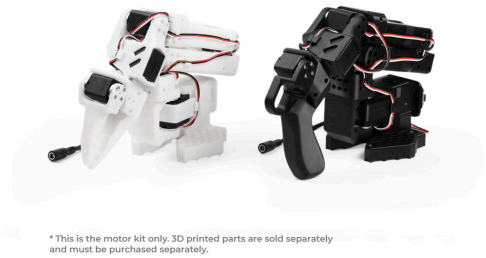
【Note】

- From China warehouse to the United States, choose "**US Direct Group Shipping**" to enjoy duty-free！
- **6 Degree-of- Freedom(DOF) Arm** — Six STS3215 magnetic-encoder bus servos deliver 30 kg·cm torque per joint with 12-bit absolute position feedback for smooth, quiet, and repeatable motion
- **Hugging Face LeRobot Native Support** — Works out of the box with the LeRobot framework for imitation learning; collect teleoperation data and train ACT/Diffusion models with community-shared pipelines.
- **Affordable & Flexible Kit Options** — Low-cost 3D-printable robotic arm with flexible kit options. Select full-assembly, servo-only pack or unassembled parts to match your budget and project needs.
- **Fully Open-Source** — All hardware designs, BOM, firmware, and software are publicly available; modify, extend, and share freely

One-Stop Guided Setup & Cloud-Powered Robot AI Training

SenseCraft Robotics is an in-house developed, exclusive platform from Seeed Studio. It delivers an all-in-one robot-AI workflow featuring guided setup, cloud training and lightweight local validation. It lets you finish essential pre-training steps — device pairing, serial binding, camera binding, calibration check and teleoperation check — all in one place, greatly streamlining your setup for education labs, R&D PoCs and early-stage robot validation projects.

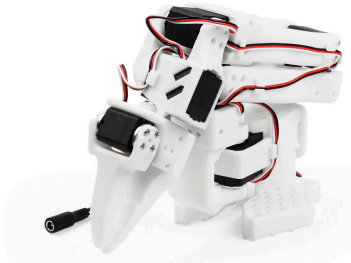
Product Gallery



* This is the motor kit only. 3D printed parts are sold separately and must be purchased separately.



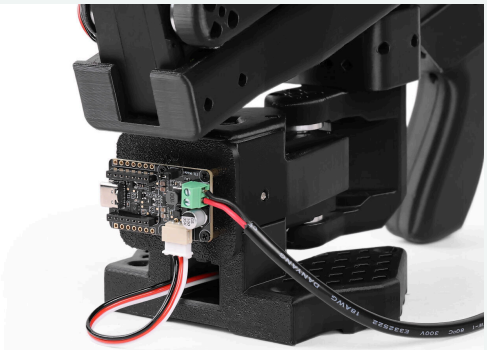
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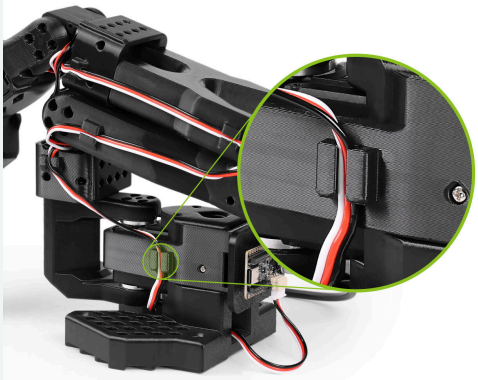
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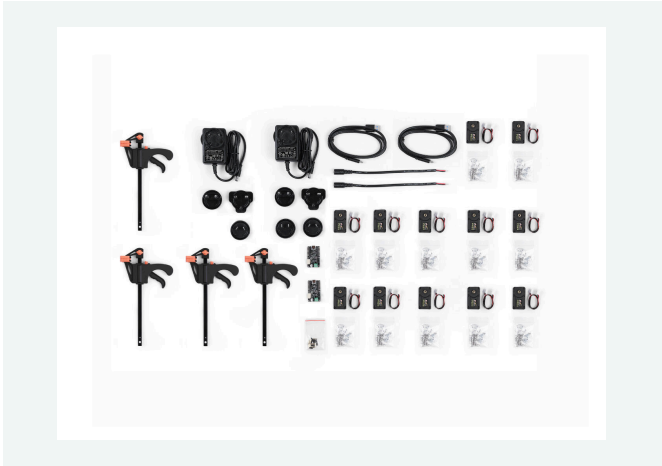


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Specification

Arm Type	SO-ARM101	
	Leader Arm	Follower Arm
Motor Configuration	1 × 7.4 V motor, 1:345 gear ratio — Joint 2	12x (12V) 1:345 gear ratio motors for all joints
	2 × 7.4 V motors, 1:191 gear ratio — Joints 1 & 3	
	3 × 7.4 V motors, 1:147 gear ratio — Joints 4 & 5 & Gripper	
Power Supply	DC 5V3A	DC 12V2A
DoF	5+1	
Payload	500g	
Angle sensor	12-bit magnetic encoder	
Operating Temperature Range	0℃～40℃	
Communication Method	UART	
Control Method	PC	

	SO-ARM101 Pro Servo Motor Kit	SO-ARM101 Pro Unassembled Bundle	SO-ARM101 Pro Kit
Price	$189.00	$295.00	$299.00
Assembly	Structural parts to be 3D-printed by user	Assembly from cratch	Pre-assembled
USB Camera	Excluded	Excluded	Included
Best For	Users with existing 3D-printed frames or custom enclosures	Hobbyists who enjoy the build process and want a complete kit. Select accessories as you like.	Researchers, educators, and teams who want to skip the build and focus on AI
Time to First Experiment	8+ hours	2-3 hours	Under 30 minutes

Applications

- **Imitation learning and Reinforcement learning:** The LeRobot framework is compatible with the SO-ARM10x robotic arm. It provides complete data collection, training, and deployment for teaching the robotic arm, and also has a comprehensive simulation platform. On the Nvidia Jetson edge device, it's easy and quick to train a customized robotic arm to complete grasping and placing tasks. For detailed tutorials, please refer to: LeRobot - NVIDIA Jetson AI Lab
- **Community Inspiration:** Walk through the whole dev journey with Nikodem Bartnik, using SO-ARM101 to showcase: from hardware assembly and servo control, to training a custom policy and achieving real-time block-picking inference. Follow his hands-on experience to understand every step, troubleshoot potential issues, and explore what's possible with this open-source robotic arm. Discover more inspiring LeRobot projects from the latest Seeed Embodied AI Hackathon!

Part List

7.4v STS3215 Servo Motor 1:345 gear rate	x1
7.4v STS3215 Servo Motor 1:191 gear rate	x2
7.4v STS3215 Servo Motor 1:147 gear rate	x3
12v STS3215 Servo Motor 1:345 gear rate	x6
Servo Driver Board for LeRobot	x2
Stud	x8
Screw	x8
5V Power Supply Cable (Multiple head)	x1
12V Power Supply Cable (Multiple head)	x1
USB-C Cable	x2
DC Power Pigtail Cable	x2
Fixing clip	x4

Compliance & Logistics



HSCODE	8501109990
USHSCODE	8501106080
UPC	
EUHSCODE	8501109390
COO	CHINA

Resource Links

- https://www.seeedstudio.com/SO-ARM101-Low-Cost-AI-Arm-Kit-p-6426.html
- STS3215 Servo Motor Datasheet
- Finedtune 3D printed parts files
- Original SO-ARM101 3D printed parts
- Servo Adaptor Board for Seeed Studio XIAO Datasheet
- reComputer J301x Datasheet
- reComputer J401X Datasheet

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